Computer Science - Graduate Level Assessment 2 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. What is the total number of colors in RGB color space?
   - (a) 16777216
   - (b) 4096
   - (c) 5000000
   - (d) 32768
   - (e) 1000000
2. The message must be encrypted at the sender site and decrypted at the
   - (a) Data center
   - (b) Sender Site
   - (c) Server site
   - (d) Conferencing
   - (e) Site
3. Let $X_0, X_1, X_2, ...bedrawni.i.d.fromp(x), andx∈ {mmdphx . . . , 100}etgwesfkyvxynwyfbnnvz
   Compute E(N)$.

   4. 99.0
   5. 101.0
   6. 125.0
   7. 150.0
   8. 50.0

Given 2 colors whose HSI representations are given as follows: (a) $(pi, 0.3,0.5)$, (b) $(0.5 pi, 0.8,0.3)

1. Brightness cannot be compared without additional information about the viewing conditions
2. (b) is brighter due to higher hue value
3. Cannot be determined from the given data
4. Brightness is subjective and cannot be determined from HSI values alone
5. (b) is brighter because it has a higher saturation and lower hue value

Which of the following is NOT a property of bitmap graphics?

1. They can support millions of colors

2. Realistic lighting and shading can be done.
3. Bitmaps can be made transparent
4. Fast hardware exists to move blocks of pixels efficiently.
5. Bitmap graphics can be created in multiple layers

True / False (10 marks)

Answer True or False

6. True or False: WPA2-Enterprise provides the strongest wireless security among commonly used
7. True or False: The function O(1) represents the smallest asymptotic growth rate among the giver
8. True or False: A heap area is usually not included in a subroutine's activation record frame in sta
9. True or False: The capacity of a cascade of n identical independent binary symmetric channels
10. True or False: Overfitting refers to a model that performs well on the training data but fails to gen

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find the list of lists with maximum length.
12. Write a function to separate and print the numbers and their position of a given string.

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